

Solution Requirements

Date	27 June 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyser
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users (Students and Professors) can securely log in using their email ID and unique password.	High
2	Authorization levels	Two roles exist: Students can record audio and view personal reports; Professors can manage batches and view collective student analytics.	High
3	External interfaces	Connects to external Speech-to-Text APIs (like OpenAI Whisper) and Large Language Model (LLM) APIs for text evaluation.	High

4	Transactions processing	System safely processes and saves audio files from the frontend into the backend file storage without data loss.	Medium
5	Reporting	Generates instant visual dashboards tracking fluency metrics, missing concepts, and a downloadable feedback PDF report.	High
6	Business rules, etc.	Audio recording length is limited to a maximum of 5 minutes per submission to maintain fair scoring.	Medium
7	Compliance to laws or regulations	Follows standard data privacy regulations by ensuring user audio logs and records are kept private and secure.	Medium
8	<i>Other</i>	System allows users to select specific technical topics (like Java, Python, or Data Structures) from a dropdown menu before starting the recording.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Processing speed for audio transcription and grading must be very efficient.	Complete report generated in < 10 seconds after submission.

2	Scalability	The application must handle simultaneous requests during peak exam hours.	Supports up to 500 concurrent users recording audio together.
3	Security & Data Privacy	All stored records, text transcripts, and credentials must be secure.	Secured using standard database access controls and secure APIs
4	Reliability & Availability	The application platform must be accessible during standard college hours.	99.5% uptime guaranteed during regular semester runs.
5	Usability & Accessibility	Simple, intuitive dashboard design suitable for both students and evaluators.	Clean UI with 3-click navigation from login to audio recording.
6	<i>Other</i>	The application must run smoothly on all major web browsers (Chrome, Edge, Safari) without requiring extra plugins.	Compatible with 100% of modern HTML5 web browsers.